Latent States in Northern Hemisphere Temperature Anomalies

Gianmaria Monaco | Bocconi University | Time Series Analysis | Group 32

1 Introduction

This analysis investigates whether Gaussian Hidden Markov Models provide a useful description of Northern Hemisphere temperature anomalies.

We first estimate models with state-dependent mean levels, both over the full sample and using observations up to 1975. We then introduce state-dependent linear trends and compare the resulting specifications using information criteria and decoded state sequences.

The states are interpreted as statistical summaries of the observed series, rather than independently established physical climate regimes.

2 Data and Seasonal Adjustment

The analysis uses monthly Northern Hemisphere temperature anomalies from GISTEMP, expressed in degrees Celsius relative to calendar-month means over the 1951–1980 reference period.

We extract the January–December observations and remove a fixed seasonal component using STL with a periodic seasonal window. Seasonal adjustment is estimated over the full sample for this retrospective analysis and re-estimated separately for the sample ending in 1975.

Since GISTEMP already reports anomalies relative to monthly reference means, STL removes only residual seasonality. The adjusted series retains the long-term movement and irregular fluctuations.

The sample contains 1752 monthly observations, from January 1880 to December 2025. The estimated seasonal component ranges from -0.033 to 0.042 °C.

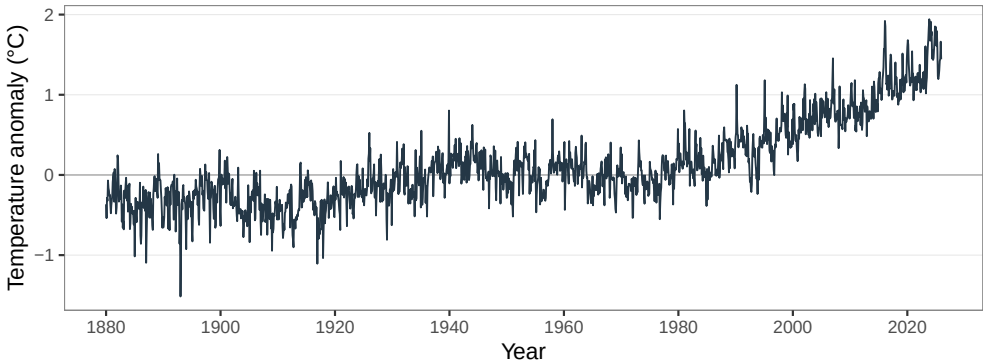


Figure 1. *Seasonally adjusted Northern Hemisphere temperature anomalies. The fixed seasonal component is estimated using periodic STL.*

3 State-Dependent Level Model

The adjusted series exhibits persistent changes in level. An HMM provides a way to summarize these patterns through latent states without fixing transition dates in advance. Estimated transition probabilities describe persistence and movement between states. This motivates its use as a descriptive model, without assuming that the states correspond to distinct physical climate regimes.

We specify a Gaussian Hidden Markov Model:

$$Y_t \mid S_t = j \sim \mathcal{N}(\mu_j, \sigma_j^2),$$

where Y_t is the seasonally adjusted temperature anomaly and S_t follows a first-order Markov chain with constant transition probabilities. Observations are assumed conditionally independent given the state sequence.

Each state has its own mean and standard deviation. We estimate models with two to five states using the EM algorithm and compare their AIC and BIC. The model with the lowest BIC is retained within this candidate set.

Each candidate model is estimated from 25 reproducible random initializations. We retain the fit with the highest finite log-likelihood and inspect its convergence status. Multiple starts reduce sensitivity to initialization but do not guarantee a global likelihood maximum.

Table 1: Comparison of state-dependent level models.

States	Log-likelihood	AIC	BIC
2	-429.89	873.78	912.06
3	-32.06	92.11	168.67
4	149.80	-253.60	-127.82
5	296.97	-525.95	-340.02

Both AIC and BIC decrease across the candidate models, reaching their lowest values with five states. We therefore retain the five-state specification for further analysis.

Since this is the largest model considered, the comparison does not establish that five states would remain preferred over a wider candidate set. Additional states may also approximate a changing temperature level, rather than identify distinct physical climate regimes. States are ordered by increasing estimated mean anomaly. The table reports the Gaussian parameters estimated by the HMM and the number of observations assigned to each state by Viterbi decoding. These counts summarize the decoded sequence.

Table 2: Estimated state parameters and decoded occupancy.

State	Mean (°C)	SD (°C)	Decoded months
1	-0.449	0.198	323
2	-0.143	0.140	566
3	0.160	0.159	472
4	0.692	0.209	258
5	1.288	0.262	133

The decoded sequence is shown below. Colored observations indicate the most likely joint state sequence, while the black step line shows the corresponding estimated state mean. Decoding uses the entire sample and therefore provides a retrospective classification.

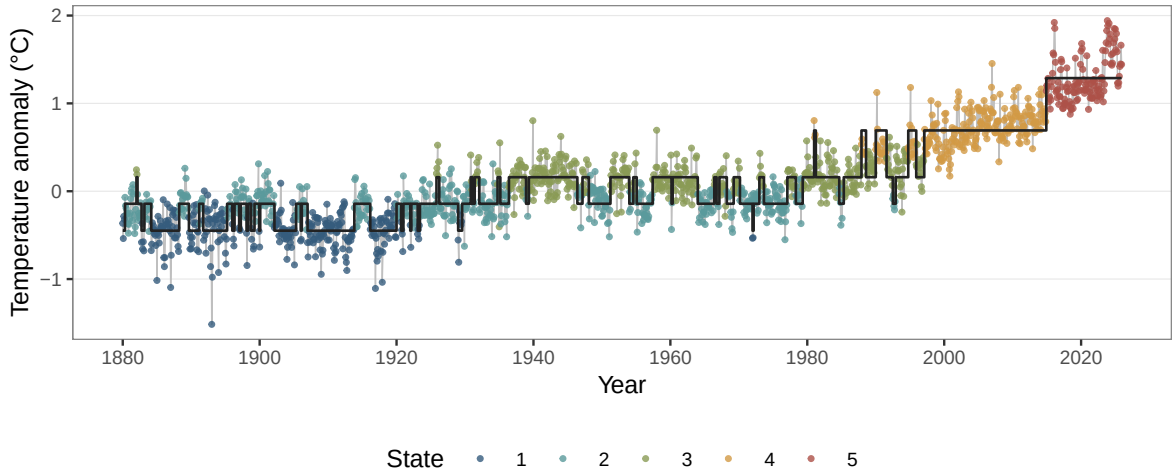


Figure 2. *Seasonally adjusted temperature anomalies colored by Viterbi-decoded state. States are ordered by increasing estimated mean. The black line follows the estimated mean of the decoded state.*

The estimated state means range from $-0.449\,^{\circ}\mathrm{C}$ to $1.288\,^{\circ}\mathrm{C}$. The earlier observations are mainly assigned to the lower-mean states, while the two highest-mean states predominantly describe the later decades.

In the five-state level model, two sustained transitions stand out in the decoded sequence: entry into State 4 in February 1997, which persists through November 2014, and entry into State 5 in December 2014, which continues to the sample end. Before 1980, the path repeatedly switches among lower-mean states. These dates summarize the fitted retrospective classification; they are not independently established physical change points.

The decoded path switches repeatedly among the lower states, whereas the recent warming is represented by sustained higher levels. This illustrates a limitation of the specification: a model with constant state means can approximate a gradual trend through successive changes of state. The decoded transitions should therefore not automatically be interpreted as abrupt physical changes in the climate system.

We examine the fitted transition probabilities to assess state persistence. Each entry gives the probability of moving from the row state to the column state in the following month. These are HMM parameter estimates, rather than transition frequencies calculated from the decoded sequence.

Table 3: Estimated monthly transition probabilities for the level model.

From	State 1	State 2	State 3	State 4	State 5
State 1	0.921	0.079	0.000	0.000	0.000
State 2	0.049	0.900	0.051	0.000	0.000
State 3	0.000	0.054	0.932	0.014	0.000
State 4	0.000	0.000	0.022	0.974	0.004
State 5	0.000	0.000	0.000	0.000	1.000

The estimated probabilities of remaining in the same state range from approximately 0.900 to 1.000. Transitions mainly occur between states with adjacent mean levels.

For the highest-mean state, the estimated probability of remaining in that state rounds to one. The decoded sequence assigns this state to the final part of the sample, with no observed departure. This limits the information available about exits from that state; the estimate should not be interpreted as evidence that the climate has entered a permanently absorbing regime.

4 Analysis up to 1975

We repeat the level-model analysis using observations from January 1880 to December 1975. Periodic STL is re-estimated on this shorter sample, excluding subsequent observations from seasonal adjustment.

We again compare models with two to five states using AIC and BIC. These criteria are used to compare specifications within this sample; their values are not directly comparable with those obtained over the full sample.

Table 4: Comparison of level models for January 1880 to December 1975.

States	Log-likelihood	AIC	BIC
2	147.86	-281.71	-246.37
3	277.99	-527.98	-457.29
4	307.35	-568.70	-452.57
5	332.48	-596.97	-425.29

Each candidate model is estimated from 25 reproducible random initializations. BIC reaches its minimum with three states, whereas AIC favors the five-state model. We retain three states according to the stated BIC selection rule. The four-state specification has a relatively close BIC, differing by 4.72 points.

Compared with the five states selected over the full sample, the shorter period supports a more parsimonious level representation under BIC. This difference is descriptive: sample length, seasonal adjustment, and the observed temperature patterns all change between the two analyses.

States are ordered by increasing estimated mean anomaly. Their labels refer to this fitted model and do not imply an exact correspondence with the states of the full-sample model.

Table 5: Estimated state parameters and decoded occupancy up to 1975.

State	Mean (°C)	SD (°C)	Decoded months
1	-0.451	0.188	312
2	-0.171	0.133	462
3	0.113	0.176	378

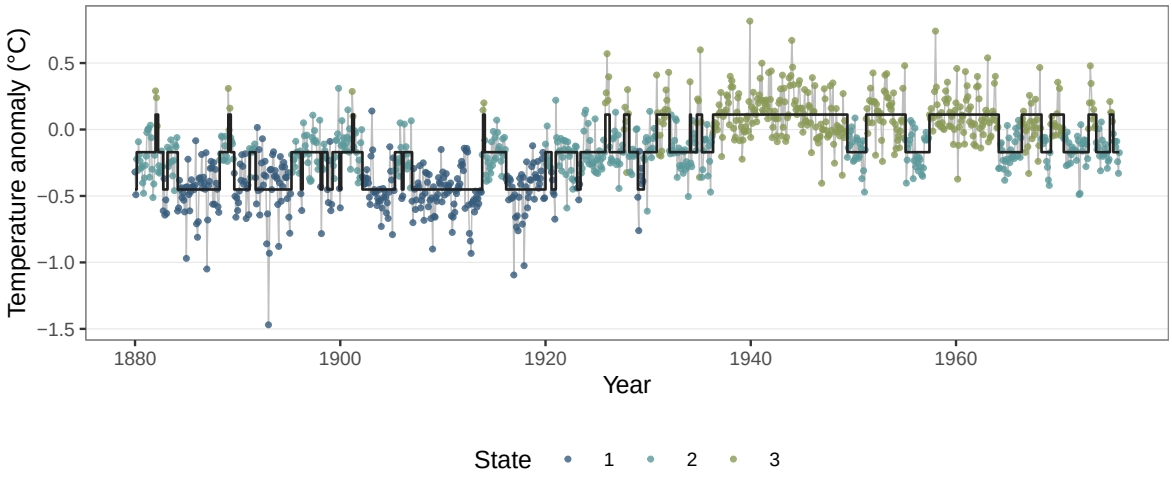


Figure 3. *Seasonally adjusted temperature anomalies up to December 1975, colored by Viterbi-decoded state. The black line follows the estimated mean of the decoded state.*

The three estimated state means range from $-0.451\,^{\circ}\mathrm{C}$ to $0.113\,^{\circ}\mathrm{C}$. The lower-mean states mainly describe the earlier decades, while the highest-mean state becomes more prominent from the 1930s onward. The decoded path also returns to lower states, rather than following a strictly increasing sequence.

Without the recent high-anomaly observations, a three-state level model provides a more compact description under BIC. However, the earlier sample still exhibits changes over time: the reduced number of states does not establish stationarity or confirm the existence of discrete physical regimes. The transition matrix below reports the estimated monthly probabilities for the shorter sample, with states ordered by increasing mean anomaly.

Table 6: Estimated monthly transition probabilities up to 1975.

From	State 1	State 2	State 3
State 1	0.932	0.068	0.000
State 2	0.047	0.904	0.049
State 3	0.000	0.058	0.942

All three states exhibit persistence, with estimated monthly self-transition probabilities ranging from 0.904 to 0.942. Unlike the full-sample model, no state has an estimated self-transition probability at the boundary value of one. However, these probabilities are comparable to those of the three lowest-mean states in the full-sample model.

Transitions mainly connect adjacent mean levels, allowing movement toward both higher and lower anomalies. Persistence alone does not establish that the states represent distinct, long-lasting physical climate regimes.

5 State-Dependent Linear Trend Model

To allow temperature anomalies to change within each state, we replace the constant state means with linear functions of time:

$$Y_t \mid S_t = j \sim \mathcal{N}\Big(\alpha_j + \beta_j(t - 1950), \sigma_j^2\Big) .$$

Here, t is calendar time in years, α_j is the state-specific expected anomaly at 1950, and β_j is the slope in degrees Celsius per year.

Transition probabilities remain constant over time. The model does not impose continuity between state-specific lines, so the decoded conditional mean may jump when states change.

We fit two- to five-state specifications to the full seasonally adjusted sample and retain the model with the lowest BIC.

Each candidate model is estimated from 150 reproducible random initializations. We retain the fit with the highest finite log-likelihood and inspect its convergence status. A global likelihood maximum is not guaranteed.

Table 7: Comparison of state-dependent linear trend models.

States	Log-likelihood	AIC	BIC
2	99.70	-181.39	-132.17
3	286.51	-539.01	-446.05
4	393.09	-732.18	-584.53
5	467.10	-856.20	-642.93

Both AIC and BIC favor the five-state trend model within the candidate set. We therefore retain five states. Since this is the largest specification considered, the comparison does not establish whether additional states would further improve the information criteria.

The selected trend model has a lower BIC than the selected level model, by approximately 302.9 points. Both models use the same observations and seasonal adjustment, so this comparison favors allowing within-state linear trends among the specifications considered. It does not establish that the decoded states correspond to physical climate regimes.

States are ordered by increasing estimated slope. The intercept represents the expected anomaly at 1950 under each state-specific line; for states mainly occupied in other periods, this is an extrapolated reference value.

The plotted conditional mean is calculated directly from the estimated HMM coefficients along the Viterbi-decoded sequence.

Table 8: Estimated parameters and decoded occupancy of the trend model.

State	Intercept (°C)	Slope (°C/year)	SD (°C)	Decoded months
1	-0.129	0.0001	0.146	294
2	-0.009	0.0063	0.154	480
3	-0.086	0.0139	0.145	361
4	0.151	0.0141	0.111	433
5	0.400	0.0166	0.175	184

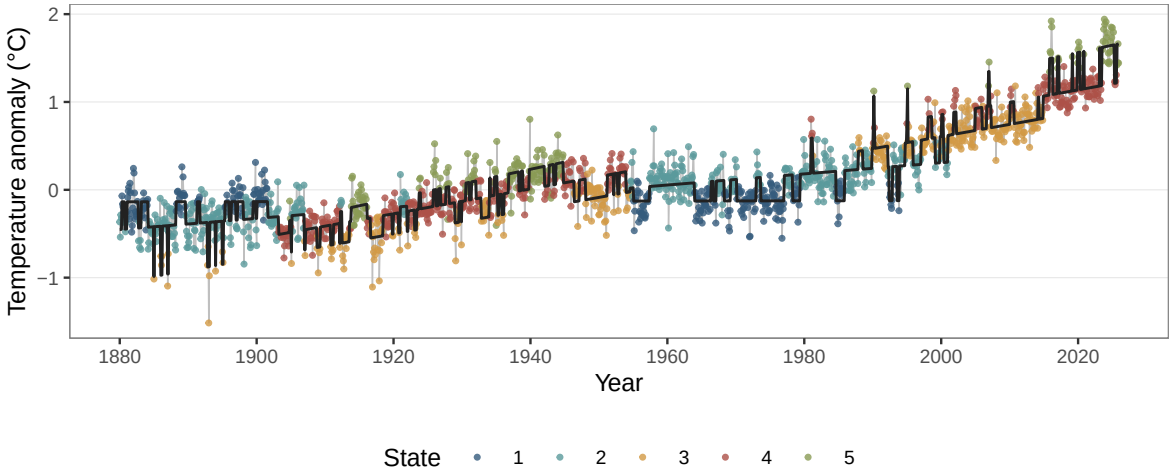


Figure 4. *Temperature anomalies colored by Viterbi-decoded state in the linear trend model. States are ordered by increasing slope. The black line follows the state-specific conditional mean along the decoded path.*

The estimated slopes range from 0.0001 to 0.0166 °C per year. State 1 has a nearly flat fitted line. States 3 and 4 have similar slopes but different intercepts, so they describe different anomaly levels despite comparable rates of change.

The decoded path revisits states across different periods. The highest-slope state appears during parts of the 1910s–1940s as well as in recent years. State labels therefore do not represent a chronological sequence of successive warming phases.

Allowing within-state trends captures gradual changes that the level model represents through discrete shifts. However, frequent switches between fitted lines still produce jumps in the conditional mean. The decoded path remains a descriptive approximation rather than evidence of discrete physical climate regimes.

6 Model Comparison

We compare the selected level and trend models with a two-state trend specification as a simpler benchmark. All models in the table use the same full sample and seasonal adjustment.

The model fitted up to 1975 is excluded from this information-criterion comparison because it uses different observations.

Table 9: Comparison of selected models and the two-state trend benchmark.

Model	States	Parameters	Log-likelihood	AIC	BIC
Selected level model	5	34	296.97	-525.95	-340.02
Selected trend model	5	39	467.10	-856.20	-642.93
Two-state trend model	2	9	99.70	-181.39	-132.17

Among the three specifications reported here, the five-state trend model has the lowest AIC and BIC. Its improvement in fitted likelihood relative to the five-state level model outweighs the penalty for its additional parameters.

The two-state trend benchmark has substantially higher information criteria. Its simpler representation is therefore less supported by this comparison. However, a two-state HMM allows repeated switches between states: it should not be interpreted as a model restricted to a single change point.

These comparisons concern in-sample fit penalized for model complexity; they do not establish forecasting performance.

The decoded two-state benchmark illustrates how the simpler specification represents the same observations. States are ordered by increasing estimated slope, and the black line follows the fitted conditional mean along the decoded sequence.

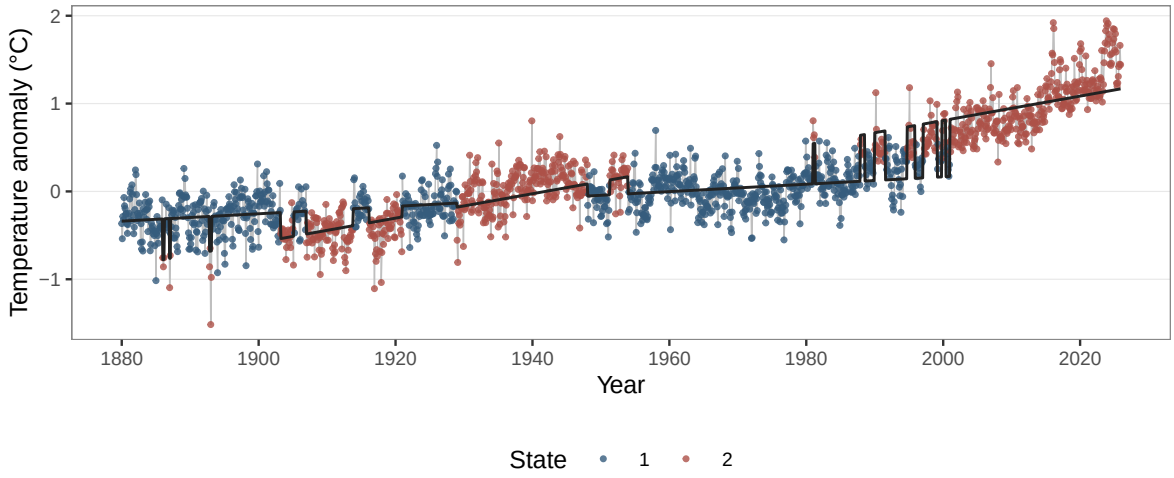


Figure 5. *Two-state linear trend benchmark: seasonally adjusted anomalies colored by Viterbi-decoded state. The black line follows the estimated conditional mean. State labels are specific to this model.*

The two-state model captures the broad increase in anomalies, but the decoded mean switches repeatedly between its two fitted lines, particularly during the 1980s and 1990s. This illustrates how a small number of state-specific trends can produce abrupt changes in the fitted path despite gradual movements in the data.

We summarize switching frequency and spell lengths in the decoded paths. We also examine lag-one autocorrelation of deviations from the decoded state-specific means. These are descriptive diagnostics: decoded states are estimated from the same observations.

Table 10: Decoded-path persistence and residual autocorrelation.

Model	State switches	Median spell (months)	Residual ACF(1)
Full-sample level	86	11	0.39
Level up to 1975	63	10	0.24
Selected trend	178	7	0.15
Two-state trend	29	24	0.55

The selected five-state trend model produces 178 decoded state switches, with a median spell length of seven months, compared with 86 switches and eleven months for the full-sample level model. Its improved information criteria therefore accompany a more fragmented decoded path, rather than a partition into a few sustained climate phases. The sample ending in 1975 also exhibits frequent switching.

Lag-one residual autocorrelation remains positive in all specifications. It is lower for the selected trend model (0.15) than for the full-sample level model (0.39) and the two-state trend benchmark (0.55). These descriptive diagnostics suggest remaining serial structure, but do not constitute a formal test of conditional independence or establish overfitting.

7 Discussion and Limitations

Gaussian Hidden Markov Models provide a descriptive representation of temperature anomalies through latent states. BIC selects five states for the full-sample level model and three for the sample ending in 1975. Allowing state-specific linear trends improves the full-sample information criteria, with both AIC and BIC favoring five states within the candidate set.

The results depend on the observation model. Constant state means approximate gradual warming through changes of level, while state-dependent trends allow movement within states but still permit discontinuities when states change. The selected trend model produces more frequent decoded switches and shorter median spells than the level model. Improved information criteria therefore do not imply a clearer partition into sustained climate regimes.

The specifications assume Gaussian observations, conditional independence given the states, and constant transition probabilities. Positive lag-one autocorrelation remains in deviations from the decoded conditional means, suggesting residual serial structure. This diagnostic does not constitute a formal test of conditional independence. Additional states may partly accommodate dependence or gradual change rather than identify distinct physical regimes.

Model selection is restricted to two through five states. Estimation uses 25 random initializations per candidate for the level models and 150 for the trend models. Multiple starts reduce sensitivity to initialization but do not guarantee a global likelihood maximum. Both selected full-sample models lie at the upper end of the candidate range. Boundary transition estimates also limit interpretation of state persistence.

Uncertainty in seasonal adjustment, parameter estimates, and state assignments is not propagated into the reported results. Estimation and decoding are retrospective: the analysis does not establish physical change points, real-time state identification, or out-of-sample forecasting performance.